

NE 511: Homework 1

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1 Problem 1

1.1 Delayed Neutron Fraction

1. To calculate the delayed neutron fractions at each of the given power levels, we need to calculate the delayed neutrons per fission and the total neutrons per fission for each material.

2. The delayed neutron fraction is given as:

$$\beta_d = \frac{\nu_d}{\nu} \quad (1)$$

3. We know the fission power is proportional to the number of fissions. Therefore, to accurately account for the different number of neutrons per fission event, we multiply the neutrons per fission by the delayed neutron fraction by the relative abundance of a given fission event (p_i) to obtain the number of delayed neutrons for each isotope. Then, sum these values together.

$$\nu_d = (\beta_{U235})(\nu_{U235})(p_{U235}) + (\beta_{Pu239})(\nu_{Pu239})(p_{Pu239}) \quad (2)$$

4. Next, we find the total ν like the ν_d , but omit scaling by delayed neutron fraction of each isotope to obtain the total number of neutrons produced from fission.

$$\nu_t = (\nu_{U235})(p_{U235}) + (\nu_{Pu239})(p_{Pu239}) \quad (3)$$

5. We then calculate the delayed neutron fraction for each state as:

$$\beta_x = \frac{\nu_{dx}}{\nu_x} \quad (4)$$

6. We also calculate the percent change in β_d from the initial state (BOC) as:

$$\%change = \frac{\beta_x - \beta_{BOC}}{\beta_{BOC}} \quad (5)$$

This gives the delayed neutron fraction for each core state:

Table 1: Delayed neutron fraction and percent change of delayed neutron fraction.

Core State	% _{power} from U-235	β_D	% change from BOC
BOC	90%	0.00598	0.0%
MOC1	85%	0.00573	-4.182%
MOC2	80%	0.00549	-8.287%
EOC	70%	0.00501	-16.271%

1.2 Core Average Neutron Lifetime

1. First, calculate the delayed group lifetimes:

$$l_{dj} = l_p + \frac{1}{\lambda_j}, \quad j = 1, \dots, 6 \quad (6)$$

2. Then, calculate the contribution from prompt neutrons for U and Pu:

$$l_{p,U235} = (1 - \beta_{U235})l_p \quad (7)$$

$$l_{p,Pu239} = (1 - \beta_{Pu239})l_p \quad (8)$$

3. Next, calculate the contribution from delayed neutrons for U and Pu:

$$l_{d,U235} = \sum_{j=1}^6 \beta_{j,U235} l_{dj} \quad (9)$$

$$l_{d,Pu239} = \sum_{j=1}^6 \beta_{j,Pu239} l_{dj} \quad (10)$$

4. We need to sum the contributions from U and Pu, but both fission types have different values for ν .

Therefore, we need to weight the contributions by the fraction of power for a given isotope p_x and the

neutrons per fission for a given isotope ν_x .

$$\frac{p_{U235}\nu_{U235}(l_{p,U235} + l_{d,U235}) + p_{Pu239}\nu_{Pu239}(l_{p,Pu239} + l_{d,Pu239})}{p_{U235}\nu_{U235} + p_{Pu239}\nu_{Pu239}} \quad (11)$$

Evaluating the neutron lifetimes for each of the operation states:

Table 2: Core Averaged Neutron Lifetimes

Core State	% _{power} from U-235	Neutron Lifetime [s]
BOC	90%	0.07886
MOC1	85%	0.07591
MOC2	80%	0.07302
EOC	70%	0.06738

2 Problem 2

For specific isotope dependence, the decay constants and concentrations are not well measured. Therefore, the isotopes with similar decay constants only have minor differences as the differences between each half-life are within the statistical error of each quantity. We group the isotopes into six groups with group yields and group half-lives. Grouping isotopes also makes kinetics calculations far easier when you only need to account for six concentrations as opposed to around a hundred individual isotopes.

3 Problem 3

1. To calculate the fraction of prompt and delayed neutrons that cause fast fission, we will first define a new group with an energy bound of 1E6 eV. Repeat the following equation for each χ .

$$\chi_{i,g} = \chi_{i,g-1} + \frac{E_{g-1} - 1e6}{E_{g-1} - E_{g+1}} (\chi_{g+1} - \chi_{g-1}) \quad (12)$$

2. Find the width of each energy group.

$$\Delta E_g = E_{g-1} - E_g \quad (13)$$

3. Interpolate all χ to be centered and not at the boundaries.

$$\bar{\chi}_{i,g} = \chi_{i,g-1} + \frac{1}{2}(\chi_{i,g} - \chi_{i,g-1}) \quad (14)$$

4. Multiply the χ 's and ΔE together.

$$f_{i,g} = \bar{\chi}_{i,g} \Delta E \quad (15)$$

5. Finally, perform this sum where the numerator only includes to groups with $E_g \geq 1MeV$.

$$f_i = \frac{\sum_{E \geq 1MeV} f_{i,g}}{\sum_{g=1}^G f_{i,g}} \quad (16)$$

For each χ_i , we obtain the following percentages:

Table 3: Percentages of neutrons in each delayed group that can cause fast fission in U-238.

χ_i	n⁰ that can cause fast fission
χ_1	1.96%
χ_2	2.65%
χ_3	1.79%
χ_{456}	9.82%
χ_p	87.92%